

assignment4 - Charges, Coulombs law 1

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:08pm

Quiz Instructions

2 attempts. I am going to make a tutorial for some of the questions. It will be found in the dropbox

REmember inverse square law. **The electric force (or electric field) decreases are the square of the distance.**

If at a distance 5cm from the source the force is 10N on a charge q then if the charge is at a distance 1 cm (divide distance by 5) then the force is $10 \times 25 = 250\text{N}$ (5 squared). if the distance is 10cm (distance is $\times 2$) then the force is $10/4\text{ N}$ (divide by 4 square of 2)



Question 1 5 pts

The force that binds atoms together to form molecules is

☐

gravitational.

☐

none of the above

☐

nuclear.

☐

electrical.

☐

centripetal.



Question 2 5 pts

Which of these does NOT have an electrical charge?

☐

proton

☐

all of the above

☐

neutron

☐

electron



none of the above



Question 3 5 pts

The force of attraction that a $-40.0 \mu\text{C}$ point charge exerts on a $+108 \mu\text{C}$ point charge has magnitude 4.00 N . How far apart are these two charges? ($k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: you are solving for the distance. I will make a tutorial. It will be found in the dropbox.



1.13 m



3.67 m



2.49 m



2.10 m



3.12 m



Question 4 5 pts

Rub electrons from your hair with a comb and the comb becomes

hint: A comb is like a balloon



positively charged.



negatively charged.



discharged.



Question 5 5 pts

Particle A has twice the charge of nearby particle B. Compared to the force on Particle A, the force on Particle B is



twice as much.



half as much.



the same.



none of the above



four times as much.



Question 6 5 pts

Two electrons are 20mm apart at closest approach. What is the magnitude of the maximum electric force that they exert on each other? ($e = 1.60 \times 10^{-19} \text{ C}$, $k = 1/4\pi\epsilon_0 = 9.0 \cdot 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

hint: convert mm to m, $q_1=q_2$ =charge of an electron. find the force. Will make a tutorial



$2.3 \times 10^{10} \text{ N}$



2.3 N



$5.8 \times 10^{-25} \text{ N}$



$5.8 \times 10^{-27} \text{ N}$



Question 7 5 pts

If 10 J of work is used in pushing 1 C of charge into an electric field, its electric potential relative to its starting position is



none of the above



10 V.



more than 10 V.



less than 10 V.



Question 8 5 pts

The electrical force on a 2-C charge is 60 N. The electric field where the charge is located is

hint: the electric field is the force per unit charge. So the force for 1 C of charge.

So $E = K (q) (1)/d^2$ with $q = 2\text{C}$. Find the force acting on 1C. this is the electric field



240 N/C.

☐

120 N/C.

☐

30 N/C.

☐

20 N/C.

☐

60 N/C.



Question 9 5 pts

Two charged particles attract each other with a force F . If the charges of both particles are doubled, and the distance between them also doubled, then the force of attraction will be

hint: use inverse square law

☐ $2 F$.☐

none of the above

☐ $F/4$.☐ F .☐ $F/2$.

Question 10 5 pts

The electrical force between charges is strongest when the charges are

☐

need more information

☐

far apart.

☐

close together.

☐

either of these



Question 11 5 pts

Two charged particles repel each other with a force F . If the charge of one of the particles is doubled and the distance between them is also doubled, then the force will be

☐ $2 F$.



$F/4$.



F .



none of the above



$F/2$.



Question 12 5 pts

Two charges separated by one meter exert 1-N forces on each other. If the charges are pulled 3 meters apart, the force on each charge will be



9 N.



3 N.



0.33 N.



0.11 N.



0 N.



Question 13 5 pts

A main difference between gravitational and electric forces is that electrical forces



attract.



act over shorter distances.



repel or attract.



obey the inverse-square law.



are weaker.



Question 14 5 pts

When the distance between two charges is halved, the electrical force between them



halves.



quadruples.



doubles.



is reduced by 1/4.



none of the above



Question 15 5 pts

An electron and a proton



attract each other.



neither attract nor repel each other.



repel each other.



Question 16 5 pts

In an electrically neutral atom the number of protons in the nucleus is equal to the number of



both electrons and neutrons.



neutrons in the nucleus.



electrons that surround the nucleus.



none of the above



Question 17 5 pts

A fundamental rule of electricity is that



unlike kinds of charges attract.



both of these



neither of these



like kinds of charges repel.



Question 18 5 pts

Which of these has the greatest mass?



neither of these

☐

both of these

☐

proton

☐

electron



Question 19 5 pts

The fundamental force underlying all chemical reactions is

☐

electrical.

☐

centripetal.

☐

nuclear.

☐

gravitational.

☐

none of the above



Question 20 5 pts

When 1.0- μC point charge is 15 m from a second point charge, the force each one experiences a force of 1.0 μN . What is the magnitude of the second charge? ($k = 1/4\pi\epsilon_0 = 9.0 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$)

☐

1.0 C

☐

25 C

☐

10 nC

☐

25 nC

☐

0.025 C

Not saved

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